LLM - 3 - oy Dars Report - 10 Dars

RLHF: Reward Model, PPO

Pretraining -> SFT -> LoRA / QLoRA ->
-> DPO -> RLHF - Bugungi mavzu

SFT - Model Instruction response namunalardan javob berishni o'rganadi
Loss: Cross - Entropy

LoRA / QLoRA - Kam parametr yangilab T4 da fine tuning qildik. QLoRA: 4 bit base model + LoRA adapter

DPO - Chosen / Rejected juftliklari orqali modelni bevosita optimallashtirdik. Reward Model ham, PPO ham kerak bo'lmadi

Alignmentning 2 asosiy yo'li

SFT dan keyin DPO ni boshlaganimizda biz allaqachon Alignment bosqichiga kirgan edik. Alignment - bitta algoritm emas, balki modelni inson hohishlariga moslashtirish jarayoni.

1-yo'l: DPO

Pretrained -> SFT -> Preference Data => DPO -> Aligned Model

2-yo'l: Traditional RLHF
Pretrained => SFT => Reward Model => PPO => Aligned Model.

Maqsad bir xil — javoblarni inson afzalliklariga moslashtirish. Yo'l har xil: DPO preference dan bevosita o'rganadi, RLHF esa avval Reward Model qurib, keyin PPO bilan optimallashtiradi.

PPO — Proximal Policy Optimization

RLHF — reinforcement learning from human feedback

Model 2 ta javob yaratadi, inson yaxshirog'ini tanlaydi. Bu tanlov modelni yaxshilash uchun signalga aylanadi.

① Human Feedback — insonning javoblar haqidagi bahosi

② Preference — bir javobning boshqasidan afzal ko'rilishi

③ Reward — javob sifatini bildiruvchi sonli signal.

④ Policy Optimization — Modelning javob yaratish strategiyasini yaxshilash.

RLHF Pipeline

Base Model
↓
Supervised Fine Tuning
↓
SFT Model
↓
Human Preference Data
↓
Reward Model Training
↓
PPO Optimization
↓
Aligned Model

1-Bosqich: Pretraining
Model katta xajmli malumot asosida o'qitiladi. Asosiy maqsad keyingi tokenni bashorat qilish

2-Bosqich: Supervised Fine Tuning. Model instructions response misollari orqali o'qitiladi. Model inson tayyorlagan namunalar asosida foydalanuvchi ko'rsatmalariga javob berishni o'rganadi.

3-Bosqich RLHF jarayonida model javoblarini insonlarning afzalliklariga moslashtirishga harakat qiladi

Human Preference Data (x, y_w / y_l)

x - Prompt foydalanuvchi savoli
y_w - chosen - afzal ko'rilgan javob (winner)
y_l - rejected - kamroq afzal javob (loser)

Xuddi shu format DPO da ham ishlatiladi. Farq - malumotdan qanday foydalanishda. DPO uni bevosita LLM ga beradi RLHF esa avval undan Reward Model o'qitadi

Preference Data Qanday yig'iladi

1- Prompt
"Explain Deep Learning to a Beginner"
2- SFT Model - Bir nechta javob yaratadi
(A, B)
3- Annotator - Javob(n) taqqoslab, yaxshini tanladi → ④ Saqlash {prompt, chosen, rejected} yoziladi

Reward Model (RM) — javobga sonli baho beruvchi model

RM javob yaratmaydi — faqat baholaydi

Ballar nisbiy: faqat taqqoslash uchun ma'noli

PPO — Proximal Policy Optimization

PPO modelni yuqoriroq reward olishga yo'naltiradi, lekin har bir yangilanishni kichik va nazorat ostida ushlaydi. Juda katta qadam — model SFT da o'rgangan foydali xususiyatlarini yo'qotadi

Old Policy — Oldingi iteratsiyadagi model

New Policy — yangilanayotgan model

Probability Ratio — yangi / eski token ehtimolligi

Advantage — Action kutilgandan qancha yaxshi

Clipping — Ratio ta'sirini [1 − ε, 1 + ε] da cheklash

Value Model — Holatning kutilgan qiymati (V(s)) — critic

Amaliyot Rejasi.

Muhit

Colab T4, transformers, trl, peft, datasets

Data

Preference Dataset: prompt / chosen / rejected

Model

Qwen 2.5 - 0.5B + 1 chiqishli head, LoRA b-n

Training

Reward Trainer - Bradley - Terry Loss

Tekshirish

Chosen va rejected ballarini taqqoslash

1-Qadam: Model va Dataset
2-Qadam: Reward Trainer b-n o'qitish
3-Qadam: Reward Modelni tekshirish

Tekshirish savollari:

1- Choosen javob yuqoriroq ball oldimi
2- Training loss kamaydimi?
3- Yangi misollarda ham to'g'ri afzal biladi
4- Noto'g'ri lekin uzun javobga yuqori ball beradimi?